

Infix to Postfix Expression Normal Method

$$7 + 5 * 3 / 5^1 + (3 - 2)$$

Infix to Postfix.

$$\Rightarrow 7 + 5 * 3 / 5^1 + 3 2 -$$

$$\Rightarrow 7 + 5 * 3 / (5^1) + (3 2 -)$$

$$\Rightarrow 7 + (5 3 *) / (5^1) + (3 2 -)$$

$$\Rightarrow 7 + (5 3 + 5^1 1) + (3 2 -)$$

$$\Rightarrow (7 5 3 * 5^1 1 +) + (3 2 -)$$

$$\Rightarrow 7 5 3 * 5^1 1 / + 3 2 - +$$

$$\Rightarrow 7 5 3 * 5^1 1 / + 3 2 - +$$

Infix to Postfix Expression

ALGORITHM

Scan the symbols of the expression from left to right and for each symbol, do the following:

- a. If symbol is an operand
 - Print that symbol onto the screen.
- b. If symbol is a left parenthesis
 - Push it on the stack.
- c. If symbol is a right parenthesis
 - Pop all the operators from the stack upto the first left parenthesis and print them on the screen.
 - Discard the left and right parentheses.
- d. If symbol is an operator
 - If the precedence of the operators in the stack are greater than or equal to the current operator, then 
Pop the operators out of the stack and print them onto the screen, and push the current operator onto the stack.
 - else
Push the current operator onto the stack.

Infix to Postfix Expression

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Ex. $7 + 5 * 3 / 5 ^ 1 + (3 - 2)$

Input	stack	postfix Output
7	-	7
+	+	7
5	+	7, 5
*	+ *	7, 5
3	+ *	7, 5, 3
/	+ /	7, 5, 3, *
5	+ /	7, 5, 3, *, 5
^	+ / ^	7 5 3 * 5
1	" pop	7 5 3 * 5 1
+	+	7 5 3 * 5 1 ^ / +
(	+ (	7 5 3 * 5 1 ^ / +
3	+ 3	7 5 3 * 5 1 ^ / + 3
-	+ (-	7 5 3 * 5 1 ^ / + 3
2	+ (-	7 5 3 * 5 1 ^ / + 3 2
)	+ pop	7 5 3 * 5 1 ^ / + 3 2 -
	+	7 5 3 * 5 1 ^ / + 3 2 - +

postfix .

Postfix Expression Evaluation

ALGORITHM

Scan the symbols of the given postfix expression from left to right and for each symbol, do the following:

1. If symbol is an operand
Push it on the stack.
2. If symbol is an operator
Pop two symbols out of the stack and apply the operator on these symbols.
Push the result on the stack
3. After scanning all the symbols of the postfix expression, pop the remaining symbol out of the stack and print it on the screen. The remaining symbol is the result obtained after evaluating the postfix expression.



$$\Rightarrow 753*51^1/+32-+$$

Postfix Expression Evaluation

<u>Symbol</u>	<u>Stack</u>	<u>Result</u>
7	7	
5	75	
3	753 pop	
*	7, 15	
5	7, 15, 5	
1	7, 15, 5, 1	
^	7, 15, 5	
/	7, 3	
+	10	
3	10, 3	
2	10, 3, 2	

$$\begin{array}{l} - \quad 10, 1 \quad 3-2=1 \\ + \quad \underline{11} \quad 10+1=11 \\ \text{Result} = 11 \end{array}$$

$$53* \Rightarrow 5 \times 3 = 15$$

$$51^1 = 5^1 = 5$$

$$15/5 = 3$$

$$7+3 = 10$$

